ThaiSLM — Architecture v8 (camera → landmarks → signs → Thai sentence → speech)

Scope: how v8 is built end to end, for reuse in the project report. Changes from v7 are marked **[v8]**. Results: `result_v8.md`. Representation space: `latent_space_v8.md`. Lab: `lab.ipynb`.

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1. System at a glance



2. Data [v8]

step	rule	effect
YouTube audit (`scripts/audit_youtube.py`)	measured cuts, people, zoom, signer size, signing-hand framing, hand visibility, label agreement with other signers; eye check on contact sheets (clip vs dictionary reference)	525 / 561 word clips kept; fingerspelled words, ceremony graphics, split screen, wrongly labelled signers removed
label normalisation (`data_prep.py v8_concept`)	"word (source dictionary)" and TSL51 politeness forms → the word	9,988 → 9,605 concepts; one class per word
vocabularies (`data_prep.py daily_vocab`)	conversation = curated list ∪ TSL51; daily = conversation ∪ common Thai words (TNC ≥ 50) signed by ≥ 2 people	327 / 1,366 words; the 9.6k "full" mode is no longer offered in the app

Everything else is unchanged from v7: MediaPipe Holistic for every video (the same `.task` file as the browser), the Standard Schema at 25 fps, left-dominant signers mirrored, held-out **signers** for evaluation, TSL51 sentence templates split tune / test.

3. Hands [v8]

Holistic finds a hand only through the pose wrist; a blurred, fast hand loses it (measured: blur, not fps, is the cause). A MediaPipe **HandLandmarker** (palm detector on the whole frame) runs only on frames where a raised, visible wrist has no hand, and each hand it finds goes to the nearer wrist (within ~1 shoulder width). Same rule in the browser (`webapp/web/lib/holistic.ts fillHands`) and on the server (`modules/mediapipe_pose.py fill_hands`). Every result carries a hand-tracking report (share of raised-hand frames with 21 points per side, fallback count, longest gap, fps).

4. Sign encoder [v8] (`modules/lmenc.py`, `scripts/train_encoder.py`)

Input features (NumPy, identical in training, ONNX runtime and the API): body frame = shoulder centre and the clip's shoulder width; per hand: 20 joints relative to the wrist ÷ hand size (handshape), wrist and centre in the body frame, centre relative to the nose, hand size (depth), seen flag; right − left hand; velocities of wrists / hand centres and handshape change; nose, shoulders, elbows, wrists. 122 numbers per frame.

Member model: Linear → 4 × [depthwise Conv1D block (k 7, GLU) + Transformer block (4 heads, explicit attention)] → frame states H [T, 256] → masked mean ‖ attention pool → MLP → 256-d L2-normalised embedding. 3.4 M parameters.

Training (local RTX 2050, ~50 min per member, resumable): CosFace (s 30, m 0.25) over 7.9k concepts + 0.5 · SupCon; batches of 64 concepts × 2 instances with a *different signer* as the positive, hard negatives re-mined from the proxies every 1,000 steps; "not a sign" class from rest, real transitions and straddling windows; augmentation = signer retargeting, mirror, speed 0.7–1.6×, affine, finger jitter, hand drop-outs, webcam frame-rate; RTMW views of the same videos as extra positives (p 0.25). Selection: mean MRR on the held-out-signer **val** sets.

Release = ensemble of three members (frame states concatenated → 768; embedding = concatenation / √3):

member	difference
v8a	the recipe above
v8b	+ EMA of weights (0.998)
v8c	+ **co-articulation context** (p 0.5: 3–10 frames of other signs glued before / after, retargeted, 1–4 frame transition) and ±30 % boundary jitter

The ensemble was chosen on the TSL51 **tune** sentences (word windows of an unseen signer in real sentences): it is the most robust to window boundaries. 10.2 M parameters in total (v7: 91.9 M), one ONNX fp32 graph of 43 MB for the whole ensemble, parity cos 1.000.

5. Continuous signing (unchanged design, retrained for v8)

Encoder once over the utterance → H → (a) BiGRU tagger on [H ‖ 13 kinematic features] → P(sign begin / inside); (b) span head distilled from the encoder's own window embeddings → embeddings of all candidate windows from H → bank scores → semi-Markov decoder → exact re-score of the chosen segments → calibrated P(correct) → transition / repeat filters → TSL word-order re-rank. **[v8]** decoder grid allows 8-frame minimum segments (fast signing), tuned separately for conversation and daily vocabularies on the tune templates with out-of-fold taggers.

6. Language and speech

Language layer unchanged (accepted words only, evidence guard, rules composer offline). **[v8]** speech = VachanaTTS (Thai VITS voices, ONNX Runtime; `.env` `TTS_PROVIDER=vachana`, `TTS_VOICE_VACHANA=th_f_1|th_f_2|th_m_1|th_m_2`), chosen with a Thai speech recogniser as judge (CER 0.022 vs 0.107 for the v7 voice), 15× faster than real time on CPU, no PyTorch in the TTS image.

7. Deployment

piece	technology
model files	`models/` : `encoder.pt` (ensemble), `onnx/encoder.onnx` + `tagger.onnx` , `span_head.npz` , `bank.npz` , `vocab_{conversation,daily}.json` , settings, `mediapipe/{holistic,hand}_landmarker.task`
runtime	`modules/runtime.py` : ONNX Runtime + NumPy (or PyTorch for development); encoder kind from `models/config.json`
service	`docker compose up --build` : gateway (nginx :8080) · web (Next.js) · sign-api (FastAPI, ONNX Runtime) · tts (VachanaTTS)
browser	MediaPipe Holistic + HandLandmarker fallback on the device; GPU light-correction filter; repeated frames skipped
diagnosis	`python main.py diagnose <video or saved request> --intended "…"` ; saved live requests carry the hand-tracking report

8. Compute and cost

All v8 training and evaluation ran on the laptop (RTX 2050 4 GB, Ryzen 5 7535HS): encoder members ~50 min each, sequence stage ~1 h 45 min, decoder tuning ~10 min. No cloud GPU job was needed this round.